

Calibration and Predictive Performance of Continuous-Time and Discrete-Time Stochastic Volatility Models in Option Pricing

ROU-JYUN YU (余柔君)

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Abstract

This study presents a comparative analysis of the Heston and GARCH(1,1) option pricing models under extreme market conditions during the COVID-19 crisis. The research covers model formulation, characteristic function derivation, parameter calibration, and empirical pricing performance evaluation. The primary objective is to investigate whether black swan events distort model calibration and affect the accuracy of option pricing. Furthermore, the study examines the capability of both models to capture volatility dynamics and approximate real market conditions during periods of severe financial turbulence. In-sample and out-of-sample analyses are conducted to evaluate pricing stability and predictive performance. The empirical findings suggest that the Heston model achieves lower and more stable pricing errors, indicating superior robustness under highly volatile market conditions.

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1 Introduction

This project evaluates how time-structural differences affect option pricing by comparing the continuous-time Heston model against the discrete-time Heston-Nandi GARCH(1,1) model. It examines whether black swan events: the COVID-19 crisis(using data:S&P 500 options,date from 2020/01/02 to 2020/03/27,total 60 transaction days), distort parameter calibration and affect the accuracy of option pricing. Furthermore, it evaluates how effectively the calibrated models approximate market conditions.

2 Data

2.1 Data processing

Using data:S&P 500 options

Date from 2020/01/02 to 2020/03/27(total 60 transaction days)

Total number of samples:89448

Step	Filtering Rule
1	Remove contracts with zero trading volume or open interest.
2	Filter by DTE: monthlies retain $3 \leq DTE \leq 180$; weeklies are not restricted.
3	Restrict moneyness to $0.8 \leq S/K \leq 1.2$.
4	Remove options with option price below 0.5.
5	Remove options violating no-arbitrage bounds: negative intrinsic value or call price exceeding spot price.
6	Compute implied volatility and discard contracts without valid IV solutions.

Table 1: Option Data Filtering Procedure

2.2 Risk-free discount rate

The risk-free interest rate r_t is approximated using the yield of the 1-year U.S. Treasury bill. This rate is used to discount expected option payoffs under the risk-neutral pricing framework.

3 Heston model

3.1 Heston dynamics

The stocks follow the dynamics:

$$dS(t) = \mu S dt + \sqrt{v(t)} S dW_1(t)$$

And the diffusion term follows the square-root mean reversion process(used by Cox , Ingersoll ,and Ross(1985))

$$dv(t) = \kappa(\theta - v(t))dt + \sigma\sqrt{v(t)}dW_2(t)$$

Where $W_1(t), W_2(t)$ are Wiener process and $Cov[W_1(t), W_2(t)] = \rho dt, \quad \rho \in [-1, 1]$

3.2 Heston PDE

Define $\tau = T - t$ as time to maturity.

$$\frac{1}{2}vS^2\frac{\partial^2 U}{\partial S^2} + \rho\sigma vS\frac{\partial^2 U}{\partial S\partial v} + \frac{1}{2}\sigma^2 v\frac{\partial^2 U}{\partial v^2} + rS\frac{\partial U}{\partial S} + [\kappa(\theta - v) - \lambda]\frac{\partial U}{\partial v} - rU - \frac{\partial U}{\partial \tau} = 0$$

With the terminal condition($\tau = 0$) :

$$U(S, v, 0) = \max(S - K, 0)$$

And boundary conditions:

$$U(0, v, \tau) = 0$$

$$\lim_{S \rightarrow \infty} \frac{\partial U}{\partial S} = 1$$

$$\lim_{v \rightarrow \infty} U(S, v, \tau) = S$$

3.3 Characteristic function

Assume an affine-form solution: $f(x, v, \tau, \varphi) = \exp(C(\tau, \varphi) + D(\tau, \varphi)v + i\varphi x)$

Differentiate and substitute into Heston pde:

$$\frac{\partial f}{\partial \tau} + \left(r - \frac{1}{2}v\right) \frac{\partial f}{\partial x} + \kappa(\theta - v) \frac{\partial f}{\partial v} + \frac{1}{2}v \frac{\partial^2 f}{\partial x^2} + \rho\sigma v \frac{\partial^2 f}{\partial x \partial v} + \frac{1}{2}\sigma^2 v \frac{\partial^2 f}{\partial v^2} = 0$$

We have

$$\left[\frac{dC}{d\tau} + ri\varphi + \kappa\theta D\right] + v \left[\frac{dD}{d\tau} - \frac{1}{2}i\varphi - \kappa D - \frac{1}{2}\varphi^2 + i\rho\sigma\varphi D + \frac{1}{2}\sigma^2 D^2\right] = 0$$

Then solve:

$$\frac{dD}{d\tau} = \frac{1}{2}\varphi^2 + \frac{1}{2}i\varphi + (\kappa - i\rho\sigma\varphi)D - \frac{1}{2}\sigma^2 D^2 \quad \text{with } D(0, \varphi) = 0, \quad \frac{dC}{d\tau} = -ri\varphi - \kappa\theta D \quad \text{with } C(0, \varphi) = 0$$

by integral on both side.

Hence, we have the coefficients: $C(\tau, \varphi), D(\tau, \varphi)$

$$C(\tau, \varphi) = ri\varphi\tau + \frac{a}{\sigma^2} \left[(b - \rho\sigma i\varphi + d)\tau - 2 \ln \left(\frac{1 - ge^{d\tau}}{1 - g} \right) \right]$$

$$D(\tau, \varphi) = \frac{b - \rho\sigma i\varphi + d}{\sigma^2} \cdot \frac{1 - e^{d\tau}}{1 - ge^{d\tau}}$$

$$\text{Where } u = -\frac{1}{2}, \quad b = \kappa + \lambda, \quad a = \kappa\theta, \quad d = \sqrt{(\rho\sigma i\varphi - b)^2 - \sigma^2(2ui\varphi - \varphi^2)}, \quad g = \frac{b - \rho\sigma i\varphi + d}{b - \rho\sigma i\varphi - d}$$

forms the characteristic function

3.4 FFT derivation

$$C(t, K) = e^{-r\tau} \int_k^\infty (e^{s_\tau} - e^k) Q(s_\tau) ds_\tau, \quad \text{where } s_\tau = \ln S_\tau, k = \ln K$$

Since $Q(s_\tau)$ is unknown, we transform into the Fourier domain via $f_\tau(\omega) = E[e^{i\omega s_\tau}]$

$$\mathcal{F}(C_{mod}(t, k)) = \int_{-\infty}^\infty e^{i\omega k} C_{mod}(t, k) dk \equiv \psi_\tau(\omega)$$

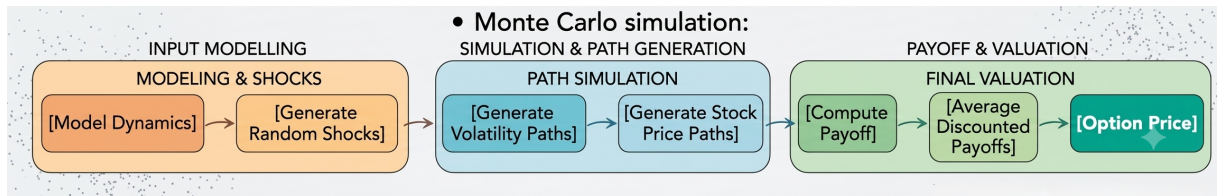
$$C_{mod}(t, k) = \mathcal{F}^{-1}(\psi_\tau(\omega)) = \frac{1}{2\pi} \int_{-\infty}^\infty e^{-i\omega k} \psi_\tau(\omega) d\omega \equiv e^{\alpha k} C(t, k), \quad \alpha > 0$$

$$\psi_\tau(\omega) = \mathcal{F}(e^{\alpha k} C(t, k)) = \int_{-\infty}^\infty e^{i\omega k} e^{\alpha k} C(t, k) dk = \frac{e^{-r\tau} f_\tau(\omega - i(\alpha + 1))}{\alpha^2 + \alpha - \omega^2 + i(2\alpha + 1)\omega}, \quad \alpha > 0$$

$$C(t, k) = \frac{e^{-\alpha k}}{2\pi} \int_{-\infty}^\infty e^{-i\omega k} \psi_\tau(\omega) d\omega = \frac{e^{-\alpha k}}{2\pi} \int_{-\infty}^\infty e^{-i\omega k} \frac{e^{-r\tau} f_\tau(\omega - i(\alpha + 1))}{\alpha^2 + \alpha - \omega^2 + i(2\alpha + 1)\omega} d\omega$$

While the FFT approach provides an efficient solution for option pricing, a simulation-based approach is essential for verifying our implementation and the consistency. Therefore, we complement the analytical method with a Monte Carlo simulation framework to construct a 95% confidence interval for the option value.

The detailed algorithm of the Monte Carlo simulation is illustrated in the flowchart below:



The process operates in three distinct stages:

- Input Modelling: Random shocks are generated based on the model's underlying stochastic dynamics.

- Path Simulation: Utilizing these shocks, we jointly simulate the asset price and volatility paths to prepare for calculating the option value.
- Payoff & Valuation: The option payoff is calculated for each individual path. According to the Law of Large Numbers, the sample average of these payoffs converges to the true expected value. The final option price is obtained by discounting this expected payoff at the risk-free interest rate r .

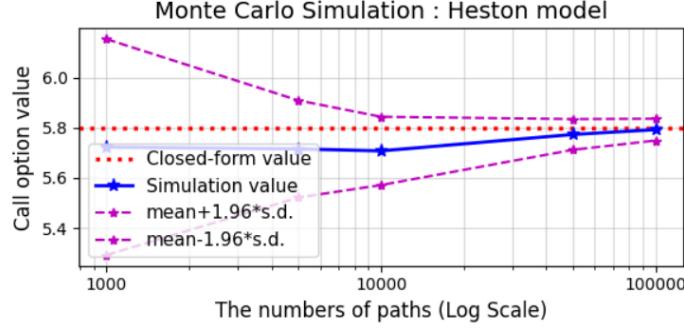


Figure 1: Heston closed form verification

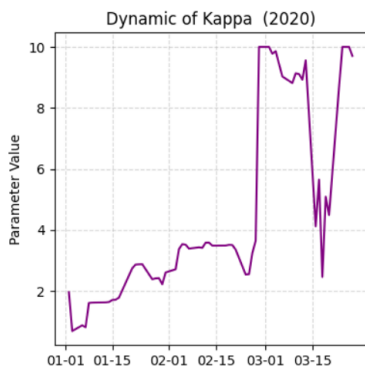
Figure 1 illustrates the convergence test results of the Monte Carlo simulation for pricing a European call option under the Heston stochastic volatility model. As shown in the figure, the red dotted horizontal line represents the theoretical closed-form solution of the Heston model (approximately 5.8). When the number of simulation paths (N) is relatively small (e.g., 1,000 paths), there is a noticeable statistical discrepancy between the simulated value (blue solid line with star markers) and the closed-form solution, accompanied by a wider 95% confidence interval (represented by the purple dashed lines). However, as the number of simulation paths increases on a logarithmic scale up to 100,000, the standard error diminishes at a rate of $O(1/\sqrt{N})$. Consequently, the 95% confidence interval narrows significantly, and the simulated value smoothly converges to the theoretical closed-form solution. The fact that the theoretical value strictly falls within the 95% confidence interval across all sample sizes verifies the correctness and statistical consistency of the implemented Monte Carlo algorithm.

3.5 Calibration results

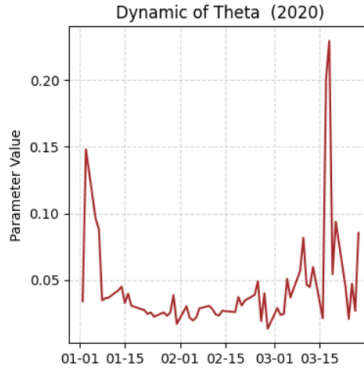
Define parameter $\Theta^H = (\kappa, \theta, \sigma, \rho, \nu_0)$, $\hat{\Theta}_t^H = \arg \min_{\Theta^H} \|\mathbf{C}_t^{\text{model}}(\Theta^H) - \mathbf{C}_t^{\text{market}}\|^2$

Parameters are calibrated by minimizing pricing errors between model and market option prices subject to the Feller condition: $2\kappa\theta > \sigma^2$ and $\rho \in [-1, 1]$

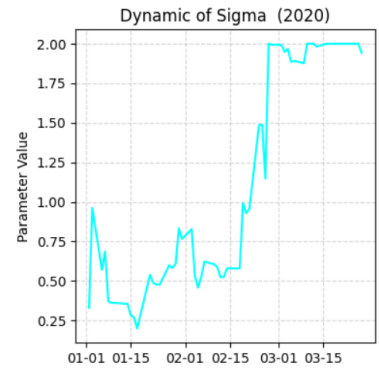
Daily dynamics of parameters:



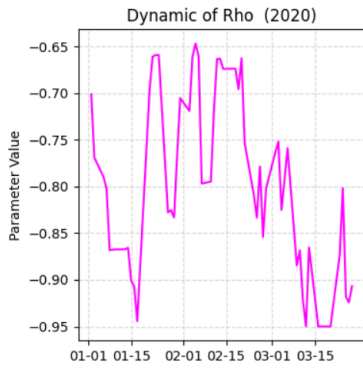
(a) Kappa



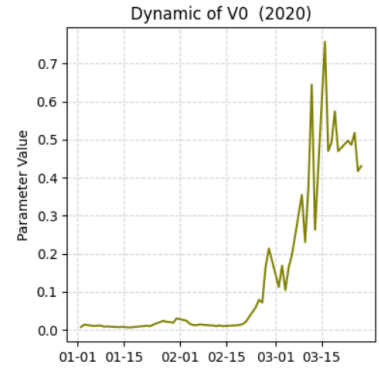
(b) Theta



(c) Sigma



(d) Rho



(e) V_0

Parameter	Meaning	Average value
κ	Mean reversion speed	4.583878
θ	Long-run variance	0.045379
σ	Volatility of volatility (Vol of Vol)	1.121638
ρ	Correlation between price and variance	-0.803435
ν_0	Initial variance	0.146251

Table 2: Heston model parameters

As shown in the calibration results. The sharp spike in v_0 and σ during March 2020 reflects the sudden explosion of market uncertainty and volatility caused by the COVID-19 crisis. The negative ρ again shows that the stock price and variance having negative covariance.

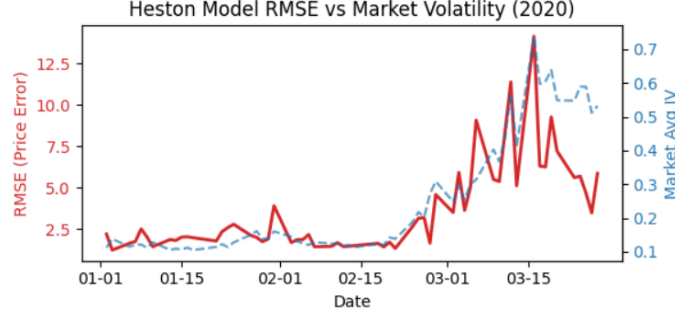


Figure 3: Heston model RMSE with market volatility

Figure 3 displays the daily calibration error (RMSE, red solid line) of the Heston model alongside the average market implied volatility (Market Avg IV, blue dashed line) during the first quarter of 2020.

The results show a strong correlation between pricing errors and market stress. From January to February, both the market volatility and the model's RMSE remain low and stable. However, during the March 2020 COVID-19 crisis, market volatility exploded from 0.1 to over 0.7. Concurrently, the pricing RMSE experienced a massive spike, peaking above 12.5. This high synchronization indicates that the Heston model struggles to accurately price options during extreme black-swan events, as continuous diffusion processes cannot fully capture sudden jumps in market volatility.

3.6 In sample performance

Mny Group	DTE Group	Mean	Std
OTM	Short	1.7212	5.1136
OTM	Mid	0.6601	3.0200
OTM	Long	-0.4526	4.1200
ATM	Short	-0.5440	4.4665
ATM	Mid	-0.4911	2.0677
ATM	Long	-0.8246	4.0314
ITM	Short	-3.3105	6.5889
ITM	Mid	-0.0469	4.2444
ITM	Long	3.6212	5.9502

Short: 0-30, Mid: 30-90, Long: >90 days to maturity

Table 3: Heston model in sample performance

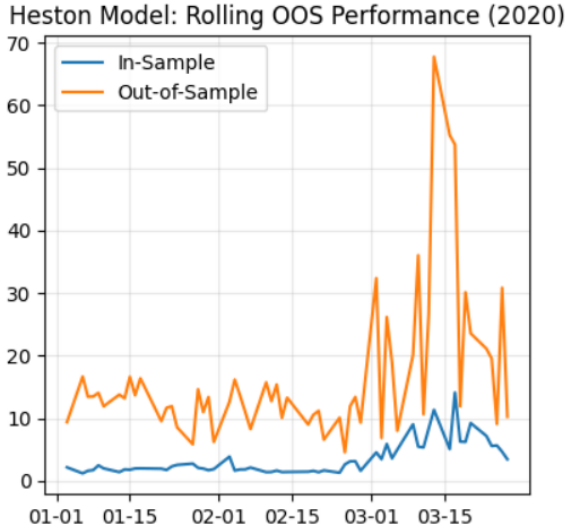
Table 3 summarizes the in-sample pricing errors of the Heston model, categorized by moneyness (Mny Group) and days-to-maturity (DTE Group). The empirical errors demonstrate distinct pricing behaviors across options groups:

- **ATM Options:** The model performs exceptionally well for At-the-Money (ATM) options, where the mean errors remain well-controlled (e.g., -0.5440 for Short DTE and -0.4911 for Mid DTE), indicating highly centered pricing accuracy.
- **OTM and ITM Options:** For Out-of-the-Money (OTM) and In-the-Money (ITM) groups, the model exhibits minor overpricing tendencies or larger standard deviations (such as a standard deviation of 6.5889 for Short ITM). This reflects the structural difficulty of capturing extreme volatility or skewness near the tails, though the overall deviations remain bounded within a statistically acceptable range.

3.7 Out of sample performance

Using yesterday’s calibrated parameters to estimate today’s option prices.

$$\text{RMSE}_{\text{OOS}} = \frac{1}{\sqrt{N_t}} \|\mathbf{C}_t^{\text{model}}(\boldsymbol{\Theta}_{t-1}) - \mathbf{C}_t^{\text{market}}\|$$



Avg IS RMSE:
3.5072

Avg OOS RMSE:
16.6363

The chart above illustrates the rolling out-of-sample pricing performance of the Heston model during the tumultuous first quarter of 2020. A glaring disparity is observed between the two domains: while the average in-Sample RMSE remains stable and remarkably low at 3.5072, the average out-of sample RMSE deteriorates sharply to 16.6363—nearly a 4.7-fold increase. This pricing gap reaches its zenith during mid-March 2020, with the OOS RMSE destabilizing and peaking near 70 as the COVID-19 financial panic unfolded.

This severe performance degradation illuminates a fundamental model misspecification and parameter fragility inherent in the standard Heston framework under stress

regimes. Because the classical Heston model relies strictly on continuous diffusion processes, it inherently lacks the mathematical machinery to accommodate price or volatility jumps. Under the historic liquidity shocks and circuit-breaking events of March 2020, the day-to-day calibration algorithm was forced to overfit the prevailing implied volatility surface by driving structural parameters to their extreme empirical boundaries. Consequently, these highly distorted parameter sets lost their intertemporal stability, leading to an immediate collapse in predictive power during out-of-sample forward extrapolation. This empirical evidence underscores the limitation of the pure-diffusion Heston model in capturing regime shifts and argues for the incorporation of jump-diffusion components (e.g., the Bates model) to handle black-swan anomalies.

4 GARCH model

4.1 GARCH dynamics

For the general GARCH(p,q)

- Stock dynamics:

$$\log(S(t)) = \log(S(t - \Delta)) + r - \frac{1}{2}h(t) + \sqrt{h(t)}z^*(t) \quad (3)$$

- Variance dynamics:

$$\begin{aligned} h(t) = \omega &+ \sum_{i=1}^p \beta_i h(t - i\Delta) + \sum_{i=2}^q \alpha_i \left(z(t - i\Delta) - \gamma_i \sqrt{h(t - i\Delta)} \right)^2 \\ &+ \alpha_1 \left(z^*(t - \Delta) - \gamma_1^* \sqrt{h(t - \Delta)} \right)^2, \end{aligned} \quad (4)$$

where, $z^*(t) = z(t) + \left(\lambda + \frac{1}{2}\right) \sqrt{h(t)}$ and $\gamma_1^* = \gamma_1 + \lambda + \frac{1}{2}$.

We use the simplest version GARCH(1,1) in this project

- Stock dynamics:

$$\ln S(t) = \ln S(t - \Delta) + r - \frac{1}{2}h(t) + \sqrt{h(t)}z^*(t)$$

- Variance dynamics:

$$h(t) = \omega + \beta_1 h(t - \Delta) + \alpha_1 (z^*(t - \Delta) - \gamma_1^* \sqrt{h(t - \Delta)})^2,$$

where $\gamma_1^* = \gamma_1 + \lambda + \frac{1}{2}$, $z^*(t) \sim N(0, 1)$

$\omega, \alpha_1, \beta_1$ are non-negative coefficients subject to the stationarity boundary constraint: $\alpha_1 \gamma_1^2 + \beta_1 < 1$.

$$\text{Cov}_{t-\Delta}[h(t + \Delta), \log(S(t))] = -2\alpha_1 \gamma_1 h(t).$$

Later we talk about GARCH model are all based on GARCH(1,1).

4.2 Characteristic function

Assume an affine-form solution:

$$f_t(\phi) = S_t^\phi \exp(A_t + B_t h_{t+1})$$

Apply backward recursion

$$f_t(\phi) = E_t[f_{t+1}(\phi)]$$

with

$$f_{t+1}(\phi) = S_{t+1}^\phi \exp(A_{t+1} + B_{t+1} h_{t+2})$$

Substitute the stock price dynamics and raise to the power ϕ
 Substitute $h_{t+2} = \omega + \beta h_{t+1} + \alpha(z_{t+1} - \gamma\sqrt{h_{t+1}})^2$ multiply by B_{t+1} .

Combine all exponential terms ,Apply the Gaussian integral formula

$$E[e^{az^2+bz}] = (1 - 2a)^{-1/2} \exp\left(\frac{b^2}{2(1 - 2a)}\right)$$

Substitute back and collect terms:

- Constant terms $\rightarrow A_t$
- Coefficients of $h_{t+1} \rightarrow B_t$

Hence, We have the recursion coefficients: A_t, B_t in the following:

$$A(t; T, \phi) = A(t + \Delta; T, \phi) + \phi r + B_1(t + \Delta; T, \phi)\omega - \frac{1}{2} \ln(1 - 2\alpha_1 B_1(t + \Delta; T, \phi))$$

$$B_1(t; T, \phi) = \phi(\lambda + \gamma_1) - \frac{1}{2}\gamma_1^2 + \beta_1 B_1(t + \Delta; T, \phi) + \frac{1/2(\phi - \gamma_1)^2}{1 - 2\alpha_1 B_1(t + \Delta; T, \phi)}$$

these coefficients can be calculated recursively from the terminal conditions:

$$A(T; T, \phi) = 0.$$

$$B_1(T; T, \phi) = 0.$$

Thus we have the generating function:

$$f(\phi) = S(t)^\phi \exp(A(t; T, \phi) + B_1(t; T, \phi)h(t + \Delta))$$

Replace ϕ to $i\phi$ is the characteristics function, can be inverted to get the risk-neutral probabilities.

4.3 Closed form derivation

Let $x = \ln(S(T))$, and $p(x)$ be the probability density

Define the moment generating function $f(\phi) = E_t[e^{\phi x}]$, $p^*(x) = \frac{e^x p(x)}{f(1)}$

then the mgf of $p^*(x)$ is $\int_{-\infty}^{\infty} e^{\phi x} p^*(x) dx = \frac{1}{f(1)} \int_{-\infty}^{\infty} e^{(\phi+1)x} p(x) dx = \frac{f(\phi+1)}{f(1)}$

$$E[\max(e^x - K, 0)] = f(1) \int_{\ln(K)}^{\infty} p^*(x) dx - K \int_{\ln(K)}^{\infty} p(x) dx$$

$$= f(1) \left(\frac{1}{2} + \frac{1}{\pi} \int_0^{\infty} \operatorname{Re} \left[\frac{K^{-i\phi} f(i\phi + 1)}{i\phi f(1)} \right] d\phi \right) - K \left(\frac{1}{2} + \frac{1}{\pi} \int_0^{\infty} \operatorname{Re} \left[\frac{K^{-i\phi} f(i\phi)}{i\phi} \right] d\phi \right)$$

$$C = e^{-r(T-t)} E[\max(e^x - K, 0)]$$

$$= \frac{1}{2} S(t) + \frac{e^{-r(T-t)}}{\pi} \int_0^{\infty} \operatorname{Re} \left[\frac{K^{-i\phi} f^*(i\phi + 1)}{i\phi} \right] d\phi - K e^{-r(T-t)} \left(\frac{1}{2} + \frac{1}{\pi} \int_0^{\infty} \operatorname{Re} \left[\frac{K^{-i\phi} f^*(i\phi)}{i\phi} \right] d\phi \right)$$

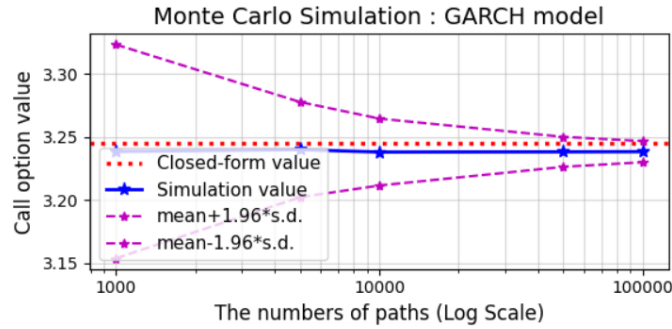


Figure 4: Garch closed form verification

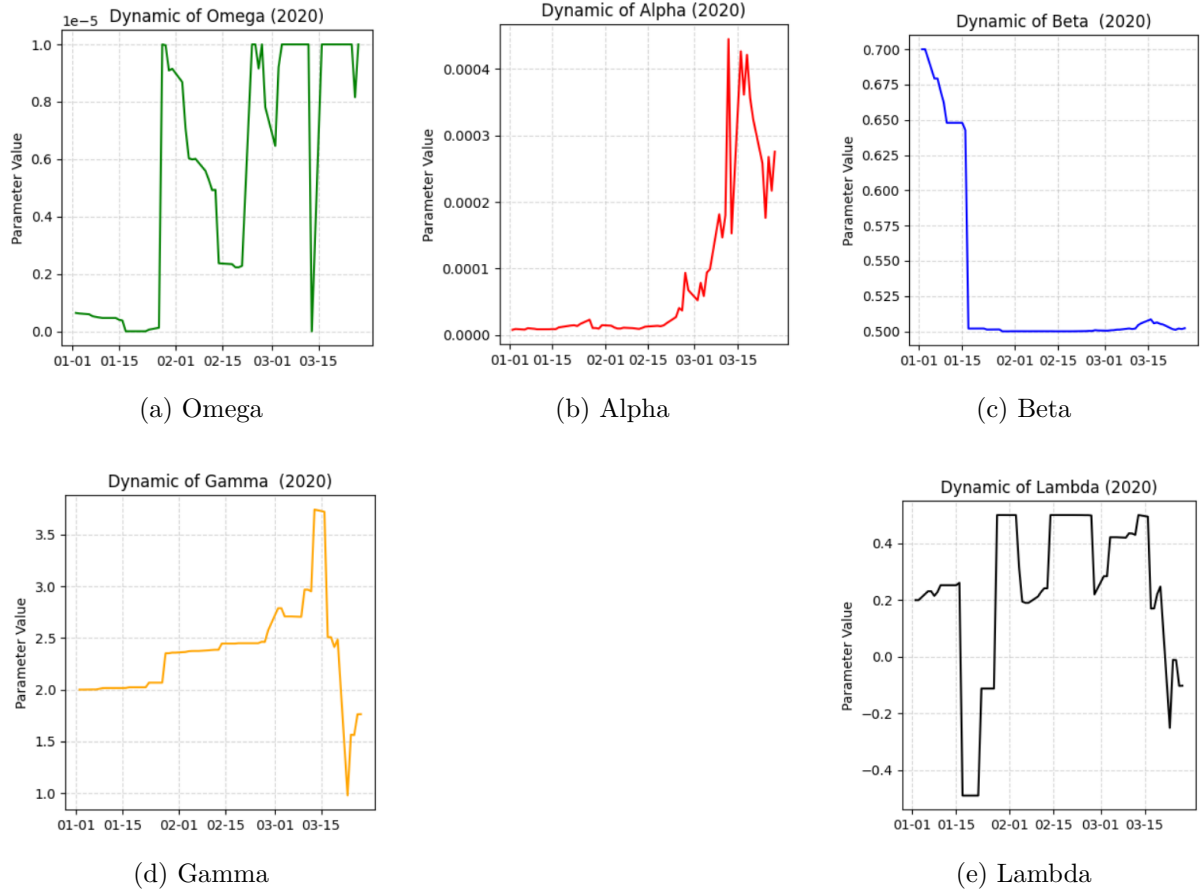
Figure 4 outlines how the Monte Carlo numerical pricing converges under the GARCH(1,1) stochastic volatility model for a European call option. The theoretical closed-form price of 3.24 is highlighted by the horizontal red dotted line. Initially, with a sparse sample size of $N = 1,000$ simulation paths, a visible statistical gap persists between the simulated output (the blue line punctuated with star markers) and the theoretical target, bounded by wide 95% confidence intervals (purple dashed lines). This estimation error is rapidly mitigated as the number of simulated trajectories expands logarithmically to 100,000. In alignment with the standard $O(1/\sqrt{N})$ error decay rate, the 95% confidence boundary narrows significantly, driving the simulated price to merge seamlessly with the exact analytical value. Shows that existence of the closed form solution.

4.4 Calibration results

Define parameter $\Theta^G = (\omega, \alpha, \beta, \gamma, \lambda)$, $\hat{\Theta}_t^G = \arg \min_{\Theta^G} \|\mathbf{C}_t^{\text{model}}(\Theta^G) - \mathbf{C}_t^{\text{market}}\|^2$

Parameters are calibrated by minimizing pricing errors between model and market option prices subject to the stationary boundary constraint: $\alpha\gamma^2 + \beta < 1$

Daily dynamic of parameters:



Parameter	Meaning	Average value
ω	Long-run baseline variance	5.68e-6
α	Sensitivity to market shocks	8.723e-5
β	Volatility persistence	0.53162926
γ	leverage effect parameter	2.33507064
λ	Risk premium parameter	0.24473856

Table 4: GARCH model parameters

The calibration results indicate that the daily dynamics of the GARCH(1,1) model parameters during the first quarter of 2020. The empirical results reveal how the model adapts to different market regimes:

- Heightened Sensitivity (α and γ): The sharp spikes in both α and γ during March 2020 show that the options market became extremely sensitive to negative shocks.

Any bad news during the COVID-19 crash immediately triggered a strong leverage effect, causing volatility to explode and cluster intensely.

- Loss of Market Memory (β): Conversely, the sudden drop in β indicates a significant decline in volatility persistence. During the peak of the panic, the market decoupled from its historical trends and was driven entirely by immediate, real-time shocks.
- Rapid Risk Repricing (ω and λ): Finally, the heavy fluctuations in ω and λ reflect the immense uncertainty in the market. As continuous circuit-breakers shook investor confidence, the baseline variance and the market price of risk were constantly being rewritten. Overall, these parameter shifts demonstrate the flexibility of the GARCH(1,1) framework in capturing the rapid transition from a stable market into a severe crisis regime.

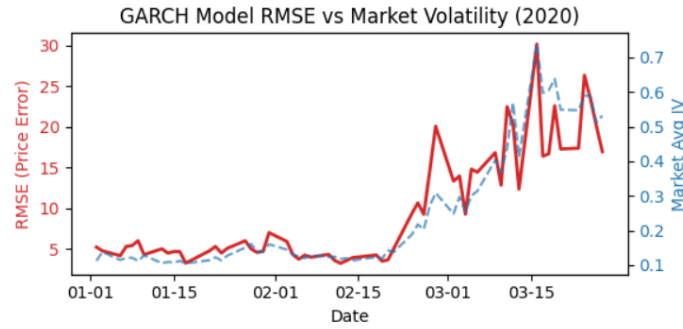


Figure 6: GARCH(1,1) model RMSE with market volatility

Figure 6 displays the daily calibration error RMSE (red solid line) of the GARCH(1,1) model alongside the average market implied volatility (blue dashed line) during the first quarter of 2020.

The results show a strong correlation between pricing errors and market stress. From January to late February, both market volatility and the model's RMSE remain low and stable. However, during the March 2020 COVID-19 crisis, as market volatility exploded from 0.1 to over 0.7, the GARCH pricing RMSE experienced a massive spike, peaking near 30. This high synchronization indicates that while the GARCH model incorporates volatility clustering, it still struggles to accurately price options during extreme black-swan events. This limitation arises because discrete-time GARCH dynamics without jump components cannot fully absorb the instantaneous and massive price shocks seen during market crashes.

4.5 In sample performance

Mny Group	DTE Group	Mean	Std
OTM	Short	-6.1101	9.1250
OTM	Mid	2.1076	8.1957
OTM	Long	10.6739	16.5945
ATM	Short	-4.7310	12.2500
ATM	Mid	-1.6494	7.1780
ATM	Long	-2.5422	6.8514
ITM	Short	-9.2454	13.7224
ITM	Mid	-4.8714	10.6095
ITM	Long	4.0709	10.8277

Short: 0-30, Mid: 30-90, Long: >90 days to maturity

Table 5: In sample performance of GARCH(1,1) model

Table 5 summarizes the in-sample pricing errors of the GARCH(1,1) model, categorized by moneyness (Mny Group) and days-to-maturity (DTE Group). The empirical results show distinct performance patterns across groups:

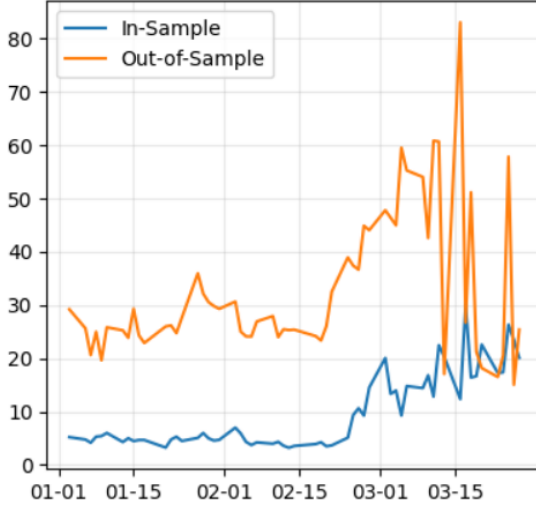
- **ATM Options:** The model performs relatively well for At-the-Money options, especially in the Mid and Long DTE categories, where mean errors remain clustered close to zero (-1.6494 and -2.5422 , respectively).
- **OTM and ITM Options:** For Out-of-the-Money and In-the-Money options, the model displays larger mean errors and higher standard deviations such as a standard deviation of 16.5945 for Long OTM. This indicates that while the GARCH framework effectively captures volatility clustering, it faces structural challenges when fitting extreme out-of-the-money tails during high-volatility regimes.

4.6 Out of sample performance

Using yesterday's calibrated parameters to estimate today's option prices.

$$\text{RMSE}_{\text{OOS}} = \frac{1}{\sqrt{N_t}} \|\mathbf{C}_t^{\text{model}}(\boldsymbol{\Theta}_{t-1}) - \mathbf{C}_t^{\text{market}}\|$$

Garch Model: Rolling OOS Performance (2020)



Avg IS RMSE:
9.5876
Avg OOS RMSE:
32.6400

The chart above displays the rolling out-of-sample performance of the GARCH(1,1) model over the same period in 2020.

Similar to the Heston model, the GARCH framework shows a noticeable split between its datasets. The average In-Sample RMSE is 9.5876, but the average OOS RMSE expands to 32.6400, peaking above 80 in mid-March. Comparing the two models, the GARCH model shows higher overall errors in both categories. This suggests that while GARCH captures volatility clustering daily, its discrete-time structure faces even greater challenges than Heston when forecasting option prices one day forward during black-swan events.

5 Model Comparison

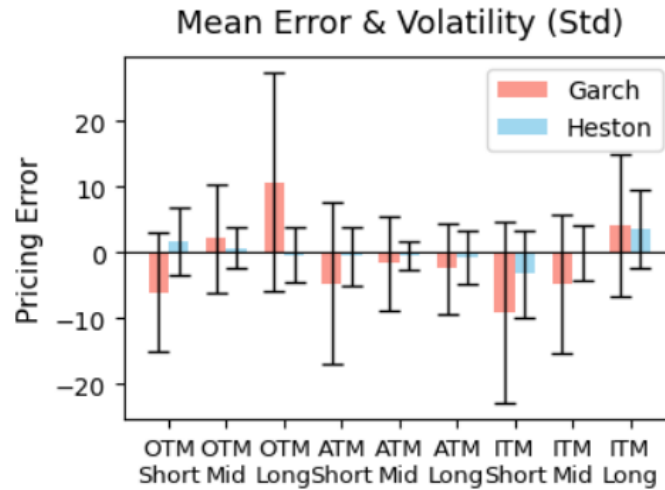


Figure 7: Different moneyness and maturity pricing error of both models

Figure 7 compares the in-sample pricing errors between the GARCH(1,1) model (red bars) and the Heston model (blue bars), categorized by moneyness and maturity groups. The colored bars represent the mean error, while the black error bars indicate the standard

deviation of the pricing errors.

The comparison yields several key insights into how both models perform under market stress:

- Overall Accuracy (Mean Error): The Heston model (blue bars) generally exhibits mean errors that are much closer to the zero baseline across most categories, particularly in the ATM and ITM groups. This indicates higher overall calibration accuracy compared to the GARCH model (red bars).
- Pricing Dispersion (Standard Deviation): The black error bars show that the GARCH model suffers from much larger standard deviations across almost all groups. For instance, in the Long OTM and Short ITM categories, the GARCH model shows massive error dispersion, whereas the Heston model keeps the error volatility significantly lower and more stable.

6 Conclusion

In this project, we have provided a comprehensive comparative analysis between the continuous-time Heston model and the discrete-time GARCH(1,1) model. Both frameworks were calibrated and tested using empirical options market data from the first quarter of 2020, capturing the unprecedented market distress triggered by the COVID-19 pandemic. The empirical findings show that the Heston model delivers more stable and consistent pricing performance under extreme market stress. As a continuous-time stochastic volatility model, Heston captures rapid shifts in market volatility more effectively than the GARCH(1,1) model. Consequently, the Heston model achieves lower pricing errors, and better approximates observed market option prices during extreme market conditions.

Despite these insights, both models experienced a noticeable deterioration in out-of-sample pricing error when market volatility exploded in mid-March 2020. This limitation suggests that pure-diffusion continuous processes and standard discrete-time GARCH dynamics struggle to fully absorb instantaneous, black-swan price shocks. Therefore, a natural extension for future research is to incorporate jump-diffusion mechanisms or embed a Markov switching model. These advanced approaches would allow the pricing kernels to dynamically adapt to sudden structural breaks, thereby enhancing forward risk management and dynamic hedging strategies during black swan events.

7 Reference

References

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